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MPJ

Assignment - 3

Problem Statement: Write a Java program to implement method overloading & method overriding in Java.

1. Write a class Shapes. Find the area of basic shapes using polymorphism (construction overloading & method overloading).

2. Write a Java program to create a parent class Hillstations & three subclasses by hill station names. Subclasses extend superclass & override methods famous food() & famousfor(). referring to parent class - Hillstations.

Objective:
1. To study method overloading
2. To study method overriding in Java.

Theory:

1. Method Overloading & its advantages

Method overloading allows multiple methods with same name but diff. parameters, improving readability, flexibility, & compile-time polymorphism.

Advantages:
- improves code readability
- reduces code duplication
- supports flexibility
- makes debugging easier
- compile-time polymorphism

2. Method Overriding & its rules

Method overriding occurs when subclass provides specific implementation of parent class method, supporting runtime polymorphism & dynamic method dispatch.

Rules:
- Method name must be same
- Parameters must be same
- Must have IS-A relationship
- Return type must be same
- Final & static methods can not be overridden.

FAQs

1. Can we overload java main() method?

Ans: Yes. By changing its parameter list. However, JVM only calls the standard main method with signature public static void main (String[] args). Others can be called manually.

2. Can we declare overloaded methods as final?

Sol^n: Yes. Final keyword prevents method overriding, not overloading. It does not affect method overloading. It prevents overriding method in a subclass.

3. Can overloaded methods be overridden?

Soln: Yes, if a subclass defines a method with same signature as one of the overloaded versions. Only those with matching parameters can be overridden in the subclass.

4. What is method overriding?

It's a feature in java where a subclass provides its own implementation of a method already defined in parent class.

5. What happens if we change arguments of overriding methods?

Soln: If the arguments of a methods are changed in a subclass, it is no longer considered method overriding but becomes method overloading. Overriding requires same method signature. No runtime polymorphism.

6. Can we change return type of overriding method from number to integer type?

Soln: Yes. we can change it to a more specific subtype, known as covariant type. This is because integer is a subclass of Number. But conversion to unrelated type is not allowed. Integer to number is also not allowed.